

Samsung SL-65A for Aerospace

Large-envelope CNC lathe with steady-rest support for precision turning of long, heavy structural round parts. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.



Big Thicket Express · 22" OD × 119" L max turning (with steady rest) · large-envelope heavy-duty CNC lathe with steady rest

Executive summary

Machine: Samsung SL-65A/3200 (shop nickname: *Big Thicket Express*). large-envelope heavy-duty CNC lathe with steady rest at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: Large-envelope CNC lathe with steady-rest support for precision turning of long, heavy structural round parts for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 22" OD × 119" L max turning (with steady rest). **Tolerance:** ±0.0005" routine on critical features.

Traceability: Material by heat and lot on every job.

The machine, in context

This is the heaviest machine on the floor. When a customer sends a 20-inch forged blank and wants it turned to a finished OD with concentric internal features across ten feet of length, Big Thicket Express is the answer. The

included steady rest is what makes long slender work possible — without it, deflection eats your tolerance.

Full spec table

Model	Samsung SL-65A/3200 ("Big Thicket Express")
Type	Large-envelope heavy-duty CNC turning center — Samsung SL-65A platform
Max turning capacity	22" OD × 119" L (B&R working spec, with steady rest)
Platform max swing	40.5" over bed
Platform max turning length	126" (3,200 mm) between centers
Chuck	24" hydraulic 3-jaw
Steady rest	Included — supports long slender work to hold straightness
Main motor	60 HP high-torque
Rigidity	Samsung SL-65 platform — heavy cast base for the largest oilfield turning work
Tolerance capability	±0.0005" routine on critical features; straightness held with steady-rest
Materials	Full range — forgings, bar stock, tubing in oilfield and structural alloys

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, ITAR handling for controlled prints, and customer-specific QMS integration.

Typical Aerospace parts we produce on Big Thicket Express

- Largest-diameter aerospace turned work
- Long forged engine and structural shafts
- Large actuator cylinder bodies
- Heavy aerospace structural components at 20"+ diameter
- Extended structural rods with steady-rest support

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

What "we run these weekly" means: we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

Customer-specific quality plans integrated on demand. First-article layout and CMM verification standard. Material traceability by heat and lot. ITAR-controlled prints handled under NDA with customer-specific document control — prints stay in-shop. Certificates of conformance issued with delivery.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys, we've handled ITAR before, and we treat first-article inspection as a discipline rather than a formality.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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